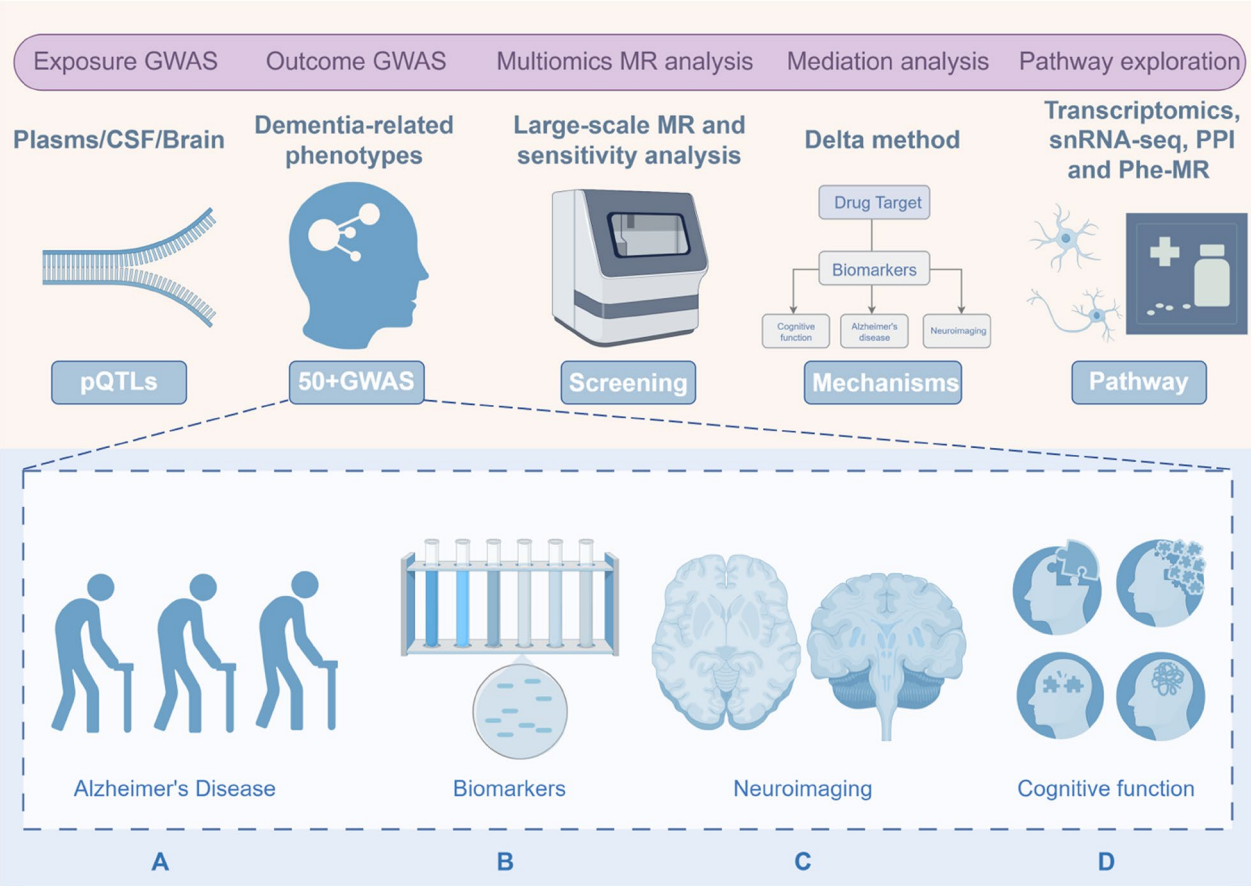


targets. PPI network analysis revealed interaction pathways between seven drug targets and four AD therapeutics, and druggability assessment revealed seven potential therapeutics.

Conclusions This study established a comprehensive AD target atlas, revealing mechanism-anchored targets that were validated across multiomics analyses and a clinically actionable framework integrating efficacy, biology, and safety profiling. Overall, these results advance AD drug discovery by revealing prioritized targets with causal biological support and providing a validated development roadmap.

Keywords Alzheimer’s disease, Multiomics causal inference, Therapeutic repurposing

Graphical Abstract



Introduction

The increasing global burden of Alzheimer’s disease (AD)—a progressive neurodegenerative disorder characterized by memory loss, cognitive decline, and pathological amyloid plaques/neurofibrillary tangles with complex multifactorial causes—poses a growing public health risk [1]. However, no disease-modifying therapies have been shown to halt AD-associated cognitive decline. The number of people in the U.S. aged 65 and older with AD is expected to skyrocket in the future; by 2060, this number could reach 13.8 million [1]. The recent conditional approval of anti-A β monoclonal antibodies (e.g., lecanemab, aducanumab), which target amyloid plaques to promote their clearance, validates the

principle of target-driven therapeutic development, and this approach provides a modest clinical benefit for some patients [2, 3]. However, the widespread application of these drugs is significantly hampered by their prohibitive costs, complex administration regimens, safety concerns (such as amyloid-related imaging abnormalities (ARIA)), and, crucially, their marginal efficacy in terms of slowing disease progression [2–4]. These limitations collectively underscore the urgent and unmet need for mechanistically diverse, accessible, and efficacious therapeutic alternatives targeting pathways for biomarkers of AD.

Human proteins serve as fundamental biological effectors and represent prime druggable targets for therapeutic intervention, as they directly mediate cellular

processes and disease pathogenesis [5]. The human proteome exhibits significant compartmentalization; proteins within the brain parenchyma and cerebrospinal fluid (CSF) are inherently more closely related to the underlying pathology of central nervous system (CNS) neurodegenerative diseases such as AD, thus reflecting the local biochemical environment and disease processes. Conversely, the peripheral blood proteome may offer a less direct view of CNS-specific mechanisms, but it is considerably more accessible for large-scale screening, diagnosis, and therapeutic monitoring, and it remains invaluable for translational applications. Recent efforts to map protein quantitative trait loci (pQTLs) across blood, CSF, and brain tissues have provided unprecedented genetic leverage for causal inference via Mendelian randomization (MR), which is a method that harnesses germline genetic variations to minimize confounding and reverse causation [6, 7]. Although proteome-wide MR has revolutionized target discovery in neurodegenerative diseases, MR studies on AD have been superficial thus far [8–10]. Prior investigations have narrowly focused on core AD clinical diagnosis as the primary endpoint, thus overlooking the critical opportunity to explore causal links to the rich landscape of preclinical AD biomarkers (e.g., A β and p-tau), detailed structural and functional neuroimaging endophenotypes (e.g., hippocampal atrophy, white matter hyperintensities, network connectivity), and the trajectories of specific cognitive domain impairments that define the clinical syndrome [10, 11].

To bridge this gap, the aim of this study was to conduct multiomics MR analyses of AD by integrating data from over 50 genome-wide association studies (GWASs) spanning AD diagnosis, CSF biomarkers (A β /p-tau), structural neuroimaging, and cognitive domains. In addition to identifying therapeutic targets, we pioneered causal mediation models that elucidated biomarker-driven

pathways, revealed target cellular hubs at single-cell resolution and performed systematic safety profiling. This comprehensive framework was designed to identify highly prioritized therapeutic targets with robust genetic evidence for causality, to provide a clear biological foundation in the context of the AD pathological cascade and to significantly reduce the risk of failure in costly clinical development owing to efficacy or safety issues.

Methods

The overall study design, which integrates pQTL and outcome GWAS data with sensitivity analyses, validation, mediation, and mechanistic exploration, is presented in Fig. 1. Details of the methodology and study participants are provided below. This study was reported in accordance with the STROBE-MR guidelines (Table S1) [12].

Study population and datasets

Proteomic data sources

We used summary-level plasma pQTL data from the deCODE consortium of Ferkingstad et al. (Table S2-S3) [5]. Additionally, pQTL data for CSF and brain tissue were obtained from Yang et al. (Table S4-S5) [7]. The results were adjusted for age, sex, and BMI to account for their potential confounding effects on protein-disease associations. We first extracted all cis-pQTLs that were strongly correlated with the corresponding exposure ($P < 5 \times 10^{-8}$). Cis-pQTLs, defined as variants located within ± 1 Mb of the encoding gene, are more likely to directly regulate protein expression and thereby reduce the risk of horizontal pleiotropy than trans-pQTLs [13]. Second, to avoid confounding effects from genomic regions with complex linkage and immune-related pleiotropy, we excluded all loci within the extended major histocompatibility complex (MHC) region (chr6: 26–34 Mb) [8]. We further applied stringent linkage disequilibrium

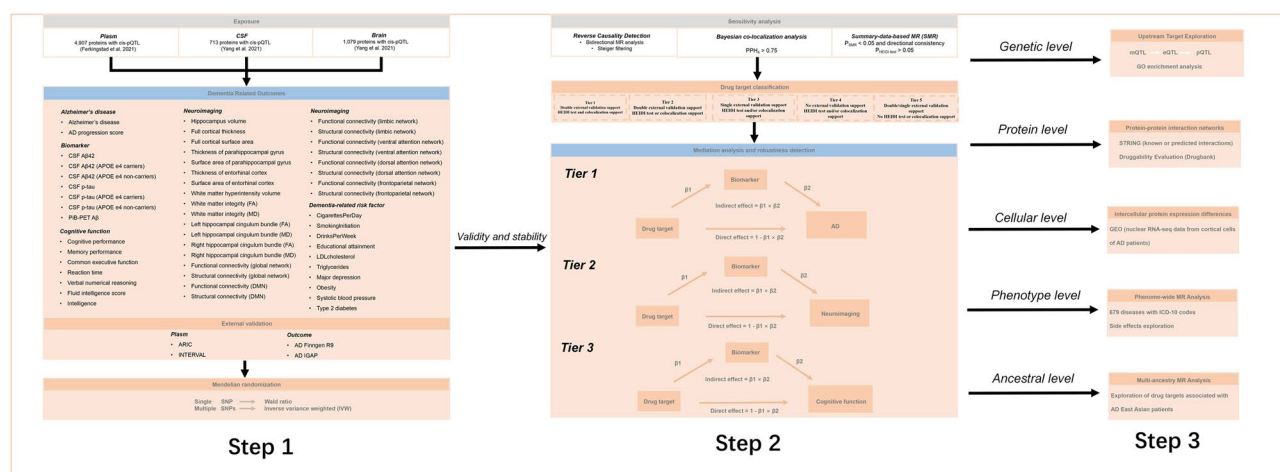


Fig. 1 Study design for the identification of plasma, CSF and brain proteins causally associated with AD, biomarkers, neuroimaging and cognitive functions

(LD) clumping ($r^2 < 0.001$ within a 10 Mb window) to ensure the independence of the instruments and reduce the inclusion of correlated variants potentially acting through shared pathways. Third, we required genome-wide significance ($P < 5 \times 10^{-8}$) to ensure robust instrument strength and further evaluated instrument validity using R^2 and F statistics. Specifically, R^2 was calculated as $R^2 = 2 \times \text{EAF} \times (1 - \text{EAF}) \times \text{beta}^2$, where EAF is the effect allele frequency and beta represents the effect size, and the F statistic was then derived as $F = R^2 \times (N - 2) / (1 - R^2)$. Variants with $F < 10$ were excluded to avoid weak or spurious instruments that might reflect confounding or pleiotropic effects. Here, R^2 represents the proportion of variance in protein levels explained by each genetic instrument. For proteins with multiple instruments, we selected those with the highest total R^2 to ensure that the strongest instruments were used for causal inference [14]. The cis-pQTLs from the Atherosclerosis Risk in Communities (ARIC) and INTERVAL cohorts served as the validation cohorts representing exposed protein drug targets [15, 16]. The ARIC cohort included 2,004 plasma proteins across 7,213 European Americans, whereas the prospective INTERVAL cohort study included approximately 50,000 blood donors aged ≥ 18 years. Plasma levels were measured via a multiplexed, aptamer-based approach (SOMAscan assay). The cis-pQTL-related genes from these two cohorts were considered for inclusion in the validation analysis.

GWAS summary statistics of outcomes

Pooled genetic data related to AD were obtained from the AD progression score [17] and 11 GWASs, with a total of 398,058 individuals of European ancestry ($n_{\text{case}}=39,918$, $n_{\text{control}}=358,140$) [18]. For external validation, genetically aggregated summary data were obtained from the International Genomics of Alzheimer's Project (IGAP) ($n_{\text{case}}=21,982$, $n_{\text{control}}=41,944$) and FinnGen ($n_{\text{case}}=9,301$, $n_{\text{control}}=367,976$) [19, 20]. To enhance the generalizability and external validity of our findings, East Asian AD GWAS ($n_{\text{case}}=3,962$; $n_{\text{control}}=4,074$) data from the NCGG biobank were used to explore drug targets among East Asian individuals [21]. This approach allowed us to investigate potential drug targets specific to East Asian individuals while acknowledging the genetic diversity across populations and exploring how ancestry may influence the relationship between genetic variation and drug effects, particularly within the framework of pharmacogenomics. Additionally, we utilized exome-wide AD age-at-onset data to perform an exploratory MR analysis [22]. Furthermore, we collected genetic summary data on 7 cognitive phenotypes (including cognitive performance, memory performance, common executive function, reaction time, verbal numerical reasoning, fluid intelligence, and intelligence, etc.) [23–26], 6 biomarkers (including

CSF A β 42, CSF p-tau, A β -PET, etc.) [17, 27], 26 neuroimaging features (including imaging features such as T1, DTI, and rsfMRI sequences) [17, 28–32], and 10 risk factors (including smoking, alcohol consumption, education level, low-density lipoprotein (LDL) cholesterol and diabetes, etc.) [23, 33–37] to allow for an integrated mechanistic inquiry into the progression of AD in terms of cognition, pathology, brain structure, and function. The reasonableness and availability of data related to AD were considered. Other relevant outcomes were selected on the basis of AD-related characteristics and pathologic features. The detailed information can be found in Table S2.

Proteome-wide Mendelian randomization (MR) analysis

In this study, we used plasma, CSF and brain tissue proteins as exposures and AD and related outcomes as endpoints. If only one pQTL was available for a given protein, the Wald ratio was used. We implemented an identical variant strategy using the same SNPs used in the primary analyses as a genetic tool and a significant variant strategy using genome-wide significant SNPs as a genetic tool to validate the primary findings. To avoid reverse causality, we performed reverse MR analysis and Steiger filtering [38]. Significant p values for the results of reverse MR analysis or Steiger filtering were considered to indicate reverse confounding. Unadjusted, raw P values are also depicted in the text. A threshold of 0.05 was used to indicate a significant association for the false discovery rate (FDR)-adjusted P value.

Bayesian colocalization analysis and summary-data-based MR (SMR) analysis

To assess whether two relevant signals (protein and risk of AD and related outcomes) were consistent with a common causal variable to distinguish confounding factors by linkage disequilibrium, we performed Bayesian colocalization analyses with summary statistics of protein and outcome GWASs [39]. The colocalization analysis included five hypotheses, the fifth of which was that intraregional SNPs share the same causal variation in exposure and outcome (PPH_4). For each protein, we included SNPs within ± 1 Mb of the pQTL. When a protein had multiple pQTLs, colocalization analyses were performed separately for each pQTL, and the pQTL with the strongest evidence of colocalization was displayed. The protein genetic locus was considered to have the same causal variant when $\text{PPH}_4 > 0.75$. While the typical threshold for colocalization is 0.8, we have applied a slightly more lenient threshold to account for the specific context of our study and the available data.

Summary data-based MR (SMR) analysis was used as a complementary method to further validate the causal relationship between proteins and AD and its associated

outcomes. The heterogeneity-dependent instrument (HEIDI) test, which uses multiple SNPs in a region, was used to distinguish proteins that are associated with outcomes due to shared genetic variation rather than genetic linkage. A p value < 0.05 was considered to indicate a significant effect for the SMR analysis. A p value > 0.05 for the HEIDI test indicated that the association of proteins with the outcome was not driven by linkage disequilibrium. Combining Bayesian colocalization and HEIDI tests can help to determine whether the causal effect of a protein target on an outcome is driven by a single locus.

Mediation analysis

Considering that the identified drug targets may affect the risk of AD by altering different biomarkers, we attempted to model the mediators to explore the possible underlying mechanisms. For proteins that were causally associated with both AD and associated traits, we performed mediation analyses to quantify the effect of the protein on AD outcomes [40]. The “total” effect of an exposure on an outcome included both the “direct” effect and any “indirect” effect through one or more mediators. In this study, total effects were captured via standard univariate MR analysis (main MR). To distinguish direct and indirect effects, we used the results of the two-step MR. The indirect effect (mediation) was estimated using the product method, where the effect of the exposure on the mediator was multiplied by the effect of the mediator on the outcome (AD-related traits). The standard error (SE) of the indirect effect was calculated using the delta method [41]. The 95% confidence interval (CI) was then derived on the basis of the SE, and the p value was calculated for the mediation effect. The mediated proportion was defined as the ratio of the mediated effect to the total effect. Considering the key role of biomarkers in the AD process, we constructed three mediator models using biomarkers ($A\beta$ and p -tau in CSF or $A\beta$ measurements via PiB-PET) as mediators. The dependent variables were AD, neuroimaging findings, and cognitive function.

Multomics analysis and single-cell type expression analysis

Given that DNA is transcribed into mRNA and that mRNA is translated into protein, we used MR and colocalization analyses to assess the concordance between protein and transcript level associations with AD. We obtained cis-expression QTL (eQTL) summary data from 31,684 blood donors and 2,856 brain tissue individuals from the eQTLGen and BrainMeta v2 databases, respectively. We selected the identified drug target genes for MR analysis [42, 43]. Additionally, we obtained brain and blood cis-methylation QTL (mQTL) summary data from the MetaBrain Consortium ($n = 1,160$) and the study by Hannon et al. ($n = 1,175$) as a supplement to further

evaluate whether the biological mechanisms of protein targets can be influenced by methylated genes [44, 45]. To investigate AD mechanisms, we performed GO enrichment analysis on the nearest genes to all significant pQTLs, with a specific focus on biological processes (BP), cellular components, and molecular functions. This analysis was conducted via the Metascape (<http://Metascape.org/gp/index.html>) web portal, with the background gene list adjusted accordingly.

We further assessed the cell type-specific expression of target genes with potential causal effects on AD at the protein locus level with Gene Expression Omnibus (GEO) single-nucleus RNA-seq data from the brain tissues of patients with Alzheimer's disease by Dai D et al. (GSE243292) [46]. To minimize potential confounding from case-control differences in cell composition, this analysis was restricted to AD samples only, thereby focusing on expression variation across different cell types within the AD context. Cells with fewer than 3 counts in a single cell and cells with fewer than 50 counts of unique features were removed. The RNA TPM was then normalized and scaled using the ‘normalize_total’ and ‘scale’ functions. To test whether the identified genes encoding AD-causing proteins were highly expressed in specific cell types of AD brain tissues, we used differential expression analysis with the Wilcoxon rank-sum test to compare the gene expression levels between one cell type and other cell types. Genes with an average log₂-fold change (log₂FC) greater than 0.25 and a FDR-adjusted P value less than 0.05 were identified as enriched genes in a particular cell type.

Protein–protein interaction (PPI), druggability evaluation and Phe-MR analysis

We explored PPI networks in plasma, CSF or brain tissue that are associated with AD risk (MR analysis $P_{FDR} < 0.05$). Specifically, we aimed to investigate the interactions between preferred proteins and whether proteins identified via plasma data could interact with proteins identified via CSF or brain tissue data. Additionally, to explore the interactions between these AD-associated genes and commercially available drug targets, we selected 3 categories of AD disease-modifying drugs from a recent review and from the DrugBank database (<https://www.drugbank.ca>) [47]. Furthermore, we examined relevant drugs with identified protein targets from the DrugBank database to explore the potential impact of clinical drugs on AD and to investigate related characteristics and the potential for repurposing drugs. Corresponding drug targets were obtained, and currently approved drugs that work against identified potential disease-causing proteins were also searched. Notably, PPI networks illustrate functional associations and do not imply that drugs targeting one protein directly modulate

interacting partners. Moreover, we incorporated 679 disease phenotypes with reference to the study by Chong et al. [48]. As in the previous study, disease outcomes were based on PheCode definitions, a system designed to organize the International Classification of Diseases and related health problem codes into phenotypic results suitable for systematic genetic analyses of numerous disease traits. Because of the lack of data availability and sufficient statistical efficacy, we excluded sex-specific related outcomes and outcomes from fewer than 500 patients. In this case, significant Phe-MR results after FDR correction were normalized to changes in biomarker levels corresponding to a 10% reduction in the risk of AD to facilitate direct comparisons of side effects (magnitude and directionality) between biomarkers. Crucially, this method inherently preserves therapeutic directionality: the sign of the normalized effect indicates whether the side effect aligns with or opposes the intended therapeutic action for AD risk modification. This established methodology is consistent with prior research [48].

Statistical analysis

When two or more genetic tools were available, inverse variance-weighted MR (MR-IVW) was applied, followed by heterogeneity analysis. The odds ratio (OR) for the risk of AD was expressed per one-standard deviation (SD) increase in plasma protein levels and per 10-fold increase in CSF and brain tissue protein levels. In the preliminary analysis, multiple tests were adjusted for further analysis using FDR-corrected P values with the Benjamini–Hochberg method. External validation was performed only for

initially identified proteins, with the p value threshold set at 0.05. MR was performed via the “TwoSampleMR” function (<https://github.com/MRCIEU/TwoSampleMR>). Bayesian colocalization analyses were performed via the “coloc” package with summary statistics of protein and outcome GWASs [39]. SMR and HEIDI tests were performed via SMR software (SMR v1.0.3) [49]. Preprocessing and transformation of the raw single-nucleus RNA-seq data were performed via the “scanpy” package in Python (version 3.8.3). The PPI analyses were performed via the Search Tool for the Retrieval of Interacting Genes (STRING) database version 11.5 (<https://string-db.org/>), with a minimum required interaction score of 0.4 [50].

Results
Proteome-wide MR analysis and sensitivity analysis

Through a range of analytical techniques, we found 15 potential drug targets for AD in GWASs among European individuals (Table 1), but these drug targets were not replicated in East Asian individuals (Table S6–S20, S43). We pooled the significant results after FDR correction and selected protein drug targets whose effects were directionally consistent with those in the SMR analysis (Table 2, S21–S26, and S32). From these, we excluded targets with potential for reverse causation. Partial exposures and outcomes with validation sets were further validated (Table S27–S29). Combining the above evidence, we categorized these proteins into five tiers (Table S30). Six proteins (PILRA isoform FDF03-M14 (plasma, OR = 0.955; 95% CI = 0.940–0.971; P_{FDR} =

Table 1 MR results for variables significantly associated with AD after FDR correction

Tissue	Protein	SNP	Effect allele	Method	OR (95% CI)	P_{FDR} value	R^2	F statistics	Target level
Plasma	PILRA isoform FDF03-M14	rs1859788	A	Wald ratio	0.955 (0.940, 0.971)	1.13E-05	0.040	1479.658	Tier 1
Plasma	PILRA isoform FDF03-deltaTM	rs1859788	A	Wald ratio	0.955 (0.940, 0.971)	1.13E-05	0.040	1479.658	Tier 1
Plasma	GRN	rs5848	T	Wald ratio	0.792 (0.716, 0.875)	1.32E-03	0.015	539.812	Tier 1
Plasma	ACE	rs4363	G	Wald ratio	0.930 (0.900, 0.960)	1.34E-03	0.040	1479.658	Tier 1
Plasma	TIMD3	rs6873659	A	Wald ratio	0.909 (0.874, 0.946)	7.28E-04	0.040	1479.658	Tier 1
Plasma	TREM2	rs75932628	T	Wald ratio	0.656 (0.583, 0.739)	2.14E-09	0.033	1213.484	Tier 1
Plasma	ACADV	rs507506	A	Wald ratio	2.465 (1.665, 3.647)	1.34E-03	0.001	37.050	Tier 3
Plasma	OMGP	rs12449302	C	Wald ratio	0.615 (0.482, 0.786)	1.49E-02	0.003	118.128	Tier 3
Plasma	BIN1	rs35103166	C	Wald ratio	1.870 (1.599, 2.188)	7.34E-12	0.007	24.351	Tier 5
CSF	Siglec-3	rs12459419	T	Wald ratio	1.511 (1.325, 1.723)	4.78E-08	0.373	496.096	Tier 2
CSF	TIMD3	rs116462788	T	Wald ratio	2.932 (1.689, 5.093)	6.08E-03	0.039	33.435	Tier 3
CSF	Siglec-7	rs2673908	A	Wald ratio	2.477 (1.511, 4.063)	8.92E-03	0.040	34.856	Tier 3
CSF	Siglec-9	rs2673908	A	Wald ratio	1.033 (1.015, 1.051)	8.92E-03	0.604	1272.118	Tier 3
Brain	APOE2	rs769449	G	Wald ratio	3.62E-05 (2.79E-05, 4.69E-05)	4.76E-322	0.166	75.227	Tier 2
Brain	IDUA	rs6599390	G	Wald ratio	0.647 (0.492, 0.850)	1.90E-02	0.175	80.081	Tier 4

Abbreviations: AD Alzheimer’s disease, CI Confidence interval, CSF Cerebrospinal fluid, MR Mendelian randomization, OR Odds ratio, SNP Single-nucleotide polymorphism (cis-acting in this research)

Table 2 Summary data for sensitivity analyses and external validation results related to possible causal proteins

Tissue	Protein	SNP	Bidirectional MR	Steiger filtering	Colocalization PPH4	P value (SMR)	HEIDI test	Validation
Plasma	PILRA isoform FDF03-M14	rs1859788	−0.002 (−0.020, 0.015)	Passed (1.14E-268)	0.993	6.79E-07	0.254	ARIC, INTERVAL, Finnngen, IGAP
Plasma	PILRA isoform FDF03-deltaTM	rs1859788	0.000 (−0.018, 0.018)	Passed (1.14E-268)	0.993	6.79E-07	0.254	ARIC, INTERVAL, Finnngen, IGAP
Plasma	GRN	rs5848	−0.019 (−0.045, 0.007)	Passed (5.06E-97)	0.991	8.05E-06	0.264	ARIC, INTERVAL, Finnngen, IGAP
Plasma	ACE	rs4363	−0.003 (−0.022, 0.017)	Passed (2.83E-273)	0.977	7.42E-06	0.674	ARIC, INTERVAL, Finnngen, IGAP
Plasma	TIMD3	rs6873659	−0.091 (−0.368, 0.186)	Passed (2.74E-272)	0.874	3.33E-04	0.706	ARIC, INTERVAL, Finnngen, IGAP
Plasma	ACADV	rs507506	0.003 (−0.017, 0.022)	Passed (5.64E-06)	0.977	2.91E-04	0.602	IGAP
Plasma	OMGP	rs12449302	−0.010 (−0.041, 0.021)	Passed (1.47E-20)	0.929	2.55E-04	0.913	IGAP
Plasma	BIN1	rs35103166	0.009 (−0.037, 0.054)	Passed (1.01E-36)	1.48E-24	2.69E-12	5.04E-07	ARIC, INTERVAL, Finnngen, IGAP
Plasma	TREM2	rs75932628	−0.150 (−0.395, 0.094)	Passed (1.43E-213)	0.922	2.21E-08	0.165	ARIC, IGAP
CSF	TIMD3	rs116462788	−0.013 (−0.125, 0.098)	Passed (2.72E-08)	0.853	1.43E-03	0.572	IGAP
CSF	Siglec-3	rs12459419	0.019 (−0.034, 0.072)	Passed (1.92E-92)	0.999	2.79E-09	2.59E-03	Finnngen, IGAP
CSF	Siglec-7	rs2673908	−0.023 (−0.059, 0.013)	Passed (1.28E-08)	0.002	1.64E-03	0.375	Finnngen
CSF	Siglec-9	rs2673908	0.010 (−0.268, 0.288)	Passed (4.72E-202)	0.002	2.69E-04	0.375	Finnngen
Brain	APOE2	rs769449	0.041 (−0.073, 0.154)	Passed (5.09E-09)	1.000	5.52E-18	7.60E-05	Finnngen, IGAP
Brain	IDUA	rs6599390	0.044 (−0.019, 0.107)	Passed (7.47E-18)	0.296	3.22E-03	0.339	NA

Abbreviations: CI Confidence interval, CSF Cerebrospinal fluid, OR Odds ratio, SMR Summary-data-based Mendelian randomization, SNP Single-nucleotide polymorphism (cis-acting in this research)

1.13×10^{-5}), PILRA isoform FDF03-deltaTM (plasma, OR=0.955; 95% CI=0.940–0.971; $P_{FDR} = 1.13 \times 10^{-5}$), GRN (plasma, OR=0.792; 95% CI=0.716–0.875; $P_{FDR} = 1.32 \times 10^{-3}$), ACE (plasma, OR=0.930; 95% CI=0.900–0.960; $P_{FDR} = 0.001$), TIMD3 (plasma, OR=0.909; 95% CI=0.874–0.946; $P_{FDR} = 7.28 \times 10^{-4}$), and TREM2 (plasma, OR=0.656; 95% CI=0.583–0.739; $P_{FDR} = 0.001$) that passed all the tests and were validated by two external datasets were categorized as Tier 1 proteins. In addition, TREM2 (plasma, OR=0.598; 95% CI=0.517–0.692; $P_{FDR} = 9.59 \times 10^{-10}$) was significantly associated with a delayed age at onset of AD. Two proteins (APOE2 (brain, OR=3.62 $\times 10^{-5}$; 95% CI=2.79 $\times 10^{-5}$ –4.69 $\times 10^{-5}$; $P_{FDR} = 4.76 \times 10^{-322}$) and Siglec-3 (CSF, OR=1.511; 95% CI=1.325–1.723; $P_{FDR} = 4.78 \times 10^{-8}$)) were categorized as Tier 2 proteins, validated by two external datasets and validated by colocalization analysis or HEIDI tests. Five proteins (TIMD3 (CSF, OR=2.932; 95% CI=1.689–5.093; $P_{FDR} = 0.006$), ACADV (plasma, OR=2.465; 95% CI=1.665–3.647; $P_{FDR} = 0.001$), OMGP (plasma, OR=0.615; 95% CI=0.482–0.786; $P_{FDR} = 0.015$), Siglec-7 (CSF, OR=2.477; 95% CI=1.511–4.063; $P_{FDR} = 0.009$), and Siglec-9 (CSF, OR=1.033; 95% CI=1.015–1.051; $P_{FDR} = 0.009$)) could be validated by an external dataset, passed colocalization analyses or HEIDI tests and were classified as Tier 3. One protein (IDUA (brain, OR=0.647; 95% CI=0.492–0.850; $P_{FDR} = 0.019$)) could not be validated by external datasets but could

be classified as Tier 4 via colocalization analysis or the HEIDI test. One protein (BIN1 (plasm, OR=1.870; 95% CI=1.599–2.188; $P_{FDR} = 0.007$)) was not validated by an external dataset, did not pass colocalization analyses or HEIDI tests and was categorized as Tier 5. Protein tiers were categorized by strength of evidence, prioritizing targets with (1) replication in two external datasets; (2) colocalization/HEIDI validation; and (3) no reverse causation. Inclusion in Tier 1 required that all the criteria were met. Inclusion in lower tiers reflected diminishing evidence. Additionally, drug targets that were significant in biomarkers, neuroimaging and cognitive function and passed colocalization analyses or HEIDI tests were also identified (Fig. 2A; Table 3). In the analysis of AD-related risk factors, plasma PILRA and ACADV were associated with changes in lipids, and ACE and BIN1 were associated with diabetes and systolic blood pressure, respectively. TIMD3 in CSF elevated plasma triglyceride levels. APOE2 in the brain was associated with LDL cholesterol and diabetes, whereas IDUA was associated with hypertension (Table S31).

Mediation analysis

We identified proteins that were causally associated with pathological markers (CSF Aβ42, CSF p-tau) related to AD outcomes (Fig. 2B, Table S33). To investigate the indirect effects of proteins on AD outcomes via pathological markers, we performed a mediation analysis

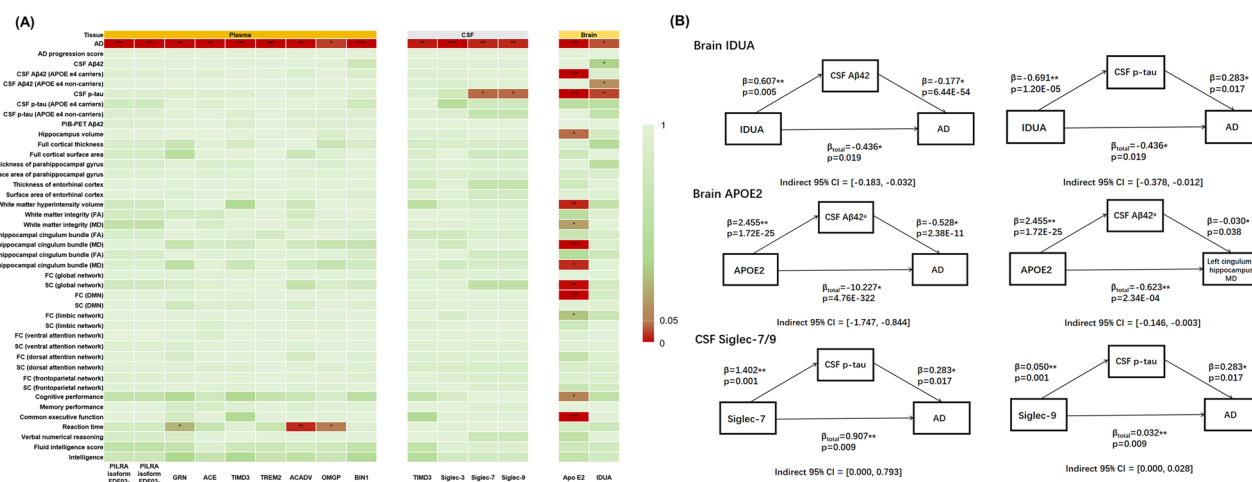


Fig. 2 A Heatmap of the significance of identified drug targets in AD, biomarkers, neuroimaging, and cognitive functions. Notes: Asterisks indicate statistically significant differences: * $P_{FDR} < 0.05$, ** $P_{FDR} < 0.01$, and *** $P_{FDR} < 0.001$. **B** Mediating effect of A β 42 and p-tau measurements on the association between identified drug targets and the risk of AD development or neuroimaging measurements. Notes: a, A β 42 measurements in CSF from APOE4-positive individuals. Notes: Asterisks indicate statistically significant differences: * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

using effect estimates from two-step MR and total effects from one MR. This analysis was limited to four proteins, i.e., APOE2 (brain), IDUA (brain), Siglec-7 (CSF), and Siglec-9 (CSF), which were found to have effects on pathological markers and AD outcomes. In brain tissue, CSF A β 42 accounted for 12.67% of the variance in the association between APOE2 and reduced AD risk, suggesting that amyloid clearance is a key protective mechanism. Additionally, CSF A β 42 accounted for 11.96% of the variance in the association between APOE2 and reduced mean diffusivity (MD) values in the cingulate gyrus tract of the hippocampus of the dominant hemisphere. Genetically predicted A β 42- and p-tau-mediated effects in the IDUA pathway accounted for 24.62% and 44.78% of the variance, respectively, in reducing AD risk, implying the involvement of dual amyloid-independent neuroprotective pathways. Additionally, p-tau accounted for 43.70% of the variance in the effects of Siglec-7 and Siglec-9 in increasing the risk of AD, highlighting tau-driven neuroinflammation as a pathogenic mechanism.

Multiomics analysis and cell type-specific expression in brain tissue

At the level of expression of accessible druggable genes, we further validated the directionality of plasma *ACADV*, *ACE*, and *TIMD3* (Table S34). The effects of *Siglec-3*, *Siglec-7*, and *Siglec-9* in CSF tissue on the risk of AD were also further clarified. At the level of gene methylation, we detected significant genetic loci in *TIMD3*, *ACE*, *ACADV*, *Siglec-3*, *Siglec-7*, *Siglec-9*, and *APOE2* (Table S35). GO enrichment analyses revealed significant convergence of prioritized AD therapeutic targets in neuroimmune pathways (Table S36). These targets were strongly enriched for macrophage and myeloid cell

activation processes in plasma ($P_{\text{FDR}} < 0.05$), involving *GRN*, *TREM2*, and *HAVCR2*, whereas CSF proteins were significantly enriched for sialic acid binding and related molecular functions ($P_{\text{FDR}} < 0.05$), driven by *Siglec* family genes (*CD33*, *Siglec-7*, and *Siglec-9*). These findings highlight dysregulated immune activation and glycoimmune checkpoints as potential shared mechanisms warranting further validation. To explore whether the genes encoding the 15 circulating proteins were enriched in the brain tissue of AD patients, we further performed single-cell type expression analysis via single-cell RNA-seq data from the GEO. The cells were clustered into 15 groups and further categorized into 7 cell types (excitatory neurons, inhibitory neurons, astrocytes, oligodendrocytes, oligodendrocyte progenitors, microglia, and endothelial cells) (Fig. 3A). Figure 3B and C show the single-cell expression of these 13 coding genes in each cluster. These protein-coding genes presented cell type-specific enrichment in the brain tissues of AD patients, with an average $\text{Log2FC} > 0.25$ and an $\text{FDR} < 0.05$ (Fig. 3D). *PILRA*, *TIMD3*, *Siglec-3*, *TREM2*, *IDUA*, *Siglec-7*, and *Siglec-9* were enriched predominantly in microglia. *GRN* was enriched mainly in endothelial cells. *APOE* was enriched mainly in astrocytes and microglia. *ACE* was enriched mainly in excitatory neurons. *BIN1* was enriched in oligodendrocytes. *OMGP* was enriched in oligodendrocyte progenitors. Additionally, our analysis was based on RNA expression rather than directly on protein expression. While RNA expression can provide valuable insights, it may not fully reflect the actual protein levels or activity.

Table 3 Summary data for sensitivity analyses and external validation results related to possible causal proteins

Phenotype	Outcome	SMR ($P_{\text{SMR}} < 0.05$, $P_{\text{HEIDI}} > 0.05$) and PPH4 > 0.75	SMR ($P_{\text{SMR}} < 0.05$, $P_{\text{HEIDI}} > 0.05$) or PPH4 > 0.75
Biomarker	CSF Aβ42	/	IDUA (brain)
	CSF Aβ42 Apo E4 positive	/	APOE2 (brain)
	CSF Aβ42 Apo E4 negative	sTie-1 (CSF)	IDUA (brain)
	CSF p-tau	/	Siglec-7 (CSF) Siglec-9 (CSF) IDUA (brain) APOE2 (brain)
	CSF p-tau Apo E4 positive	/	/
	CSF p-tau Apo E4 negative	/	/
	PiB-PET Aβ	/	/
Brain structure	Hippocampus volume	/	CPNE1 (brain) APOE2 (brain)
	Full thickness	PDLI4 (plasma)	/
	Full surface area	/	/
	Parahippocampus thickness	/	LRP8 (CSF)
	Parahippocampus surface area	/	Met (plasma) NADK (plasma)
	Entorhinal thickness	/	/
	Entorhinal surface area	/	/
	White matter hyperintensity volume	/	PPAC (CSF) TFPI (CSF) PPAC (brain) APOE2 (brain)
	White matter integrity FA	/	PGCB (plasma)
	White matter integrity MD	/	APOE2 (brain)
	Right cingulum-hippocampus FA	/	/
	Left cingulum-hippocampus FA	/	/
	Right cingulum-hippocampus MD	APOE2 (brain)	/
	Left cingulum-hippocampus MD	PGCB (plasma) SARP-2 (plasma) sTie-1 (CSF)	APOE2 (CSF) APOE2 (brain)
	Global functional connectivity	/	/
	Global structural connectivity	ERBB1 (CSF)	APOE2 (brain) sTie-1 (CSF)
	Default mode network functional connectivity	/	APOE2 (brain) MSP (brain)
	Default mode network structural connectivity	/	/

Table 3 (continued)

Phenotype	Outcome	SMR ($P_{\text{SMR}} < 0.05$, $P_{\text{HEIDI}} > 0.05$ and $\text{PPH4} > 0.75$)	SMR ($P_{\text{SMR}} < 0.05$, $P_{\text{HEIDI}} > 0.05$ or $\text{PPH4} > 0.75$)
Cognitive function	Fluid intelligence score	FOXO3A (plasma) FKBP2 (plasma) RCL (plasma) DJB12 (plasma) KS6A1 (plasma)	HEXI1 (plasma) HEXI2 (plasma) CACP (plasma) C1-Esterase Inhibitor (plasma) MSP (brain) CPNE1 (brain)
	Intelligence	FOXO3A (plasma) FKBP2 (plasma) DJB12 (plasma) PCYOX (plasma) RMD1 (plasma) KS6A1 (plasma) ENPP5 (plasma) BR serine/threonine kinase 2 (plasma)	RAB21 (plasma) HEXI2 (plasma) EPHB1 (plasma) PSME1 (plasma) FHR3 (plasma), TIMP-4 (plasma) NADPH-P450 Oxidoreductase (plasma) TCEA2 (plasma) sTie-1 (CSF) LRP8 (CSF) sICAM-1 (CSF) CPNE1 (brain) ARTS1 (brain)
	Cognitive performance	FOXO3A (plasma) FKBP2 (plasma) PCYOX (plasma) DJB12 (plasma) ENPP5 (plasma) RAB21 (plasma)	IRF-3 (plasma) HEXI2 (plasma) EFN2 (plasma) KS6A1 (plasma) C1-Esterase Inhibitor (plasma) RMD1 (plasma) PSME1 (plasma) ARLY (plasma) NAD(P)H dehydrogenase (plasma) ST4S6 (plasma) RCL (plasma) EPHB1 (plasma) TCEA2 (plasma) TXN4B (plasma) ESAM (plasma) FHR3 (plasma) BR serine/threonine kinase 2 (plasma) TIMP-4 (plasma) sTie-1 (CSF) LRP8 (CSF) CBPE (CSF) CPNE1 (brain) ARTS1 (brain) APOE2 (brain) SIG14 (brain)
	Prospective memory task	/	LRP8 (CSF) NAGK (CSF)
	Common execution functions	RCL (plasma) CJ058 (plasma) SYDC (plasma) CPLX1 (plasma) FOXO3A (plasma)	F172A (plasma) LPH (plasma) KCAB2 (plasma) PTK7 (plasma) TBCE (plasma) D3D2 (plasma) FKBP2 (plasma) AP4A (plasma) PLXB2 (plasma) SYDM (plasma) SF3B4 (plasma) RMD1 (plasma) IRF-3 (plasma) MYOC (plasma) NRP1 (plasma) ARLY (plasma) ADAMTS-5 (plasma) Growth hormone receptor (CSF) APOE2 (brain) ARTS1 (brain)
	Reaction time	FOXO3A (plasma) MXRA7 (plasma) DNJB4 (plasma) Lysozyme (plasma)	CA2D3 (plasma) ENTP6 (plasma) ACADV (plasma) F172A (plasma) Thrombin (plasma) FKBP2 (plasma) ECM1 (plasma) SHAN3 (plasma) OMGP (plasma) ECM1 (CSF)
	Verbal numerical reasoning	/	/

Abbreviations: CSF Cerebrospinal fluid, SMR Summary-data-based Mendelian randomization

PPI network analysis, Phe-MR analysis and druggability evaluation

Our PPI network revealed direct interactions between seven prioritized proteins and four targets of current therapeutic drugs for AD (Fig. 3E, Table S37-S38). Specifically, we found that BIN1 and TREM2 are associated with APP via the use of STRING. ACE-ACHE, GRN-APP, and TIMD3-IL1B may be coexpressed according to text mining. Siglec-3 interacts with IL1B and APP; and APOE2 is widely associated with APP, ACHE, BCHE, IL1B, NOX4, and NOS1. Additionally, TIMD3 was found to have extensive associations with Siglec isoforms, suggesting that these two types of proteins may be in close proximity but not necessarily in direct contact.

Our normalized Phe-MR results suggest that the identified drug targets have a complex pattern of effects on specific diseases (Table S39-S42). ACADV may increase the risk of developing nonepithelial cancer of the skin. PILRA may increase the risk of retinal detachment and symptoms concerning nutrition, metabolism, and development. We found no markedly adverse effects of other drug targets. We also searched the DrugBank database for drugs currently targeting the identified potential causal proteins. Seven potentially disease-modifying drugs were identified: perindopril, rescinamine and captopril (3 ACE inhibitors), gemtuzumab ozogamicin

and AVE9633 (2 Siglec-3 regulators), N-acetyl-alpha-D-glucosamine (1 Siglec-7 regulator), and chondroitin sulfate (1 IDUA substrate). Further longitudinal studies examining the effects of these drugs will help to clarify the pathogenesis and guide the treatment of AD.

Discussion

Identifying possible mechanisms for drug targets in AD is a major goal of genetic research. We performed a large-scale and multiomics MR analysis of AD and its associated characteristics. We identified 15 candidate drug targets and categorized them into five tiers on the basis of the strength of genetic evidence, mechanistic evidence and validation across multiple datasets. To demonstrate this tiering strategy, we highlight established AD risk loci such as TREM2 (plasma) and APOE2 (brain), categorized as Tier 1 and 2, respectively, on the basis of consistent replication in two external datasets and strong mechanistic support, which were captured by our framework. Similarly, Siglec-7/9 (CSF), whose inclusion underscores a potential mechanism related to CSF p-tau, exhibited a significant risk effect on AD progression and strongly colocalized with AD phenotypes, and enrichment analyses further indicated its involvement in sialylated glycan and organic acid binding pathways. This finding demonstrates that our tiering strategy not only recapitulates

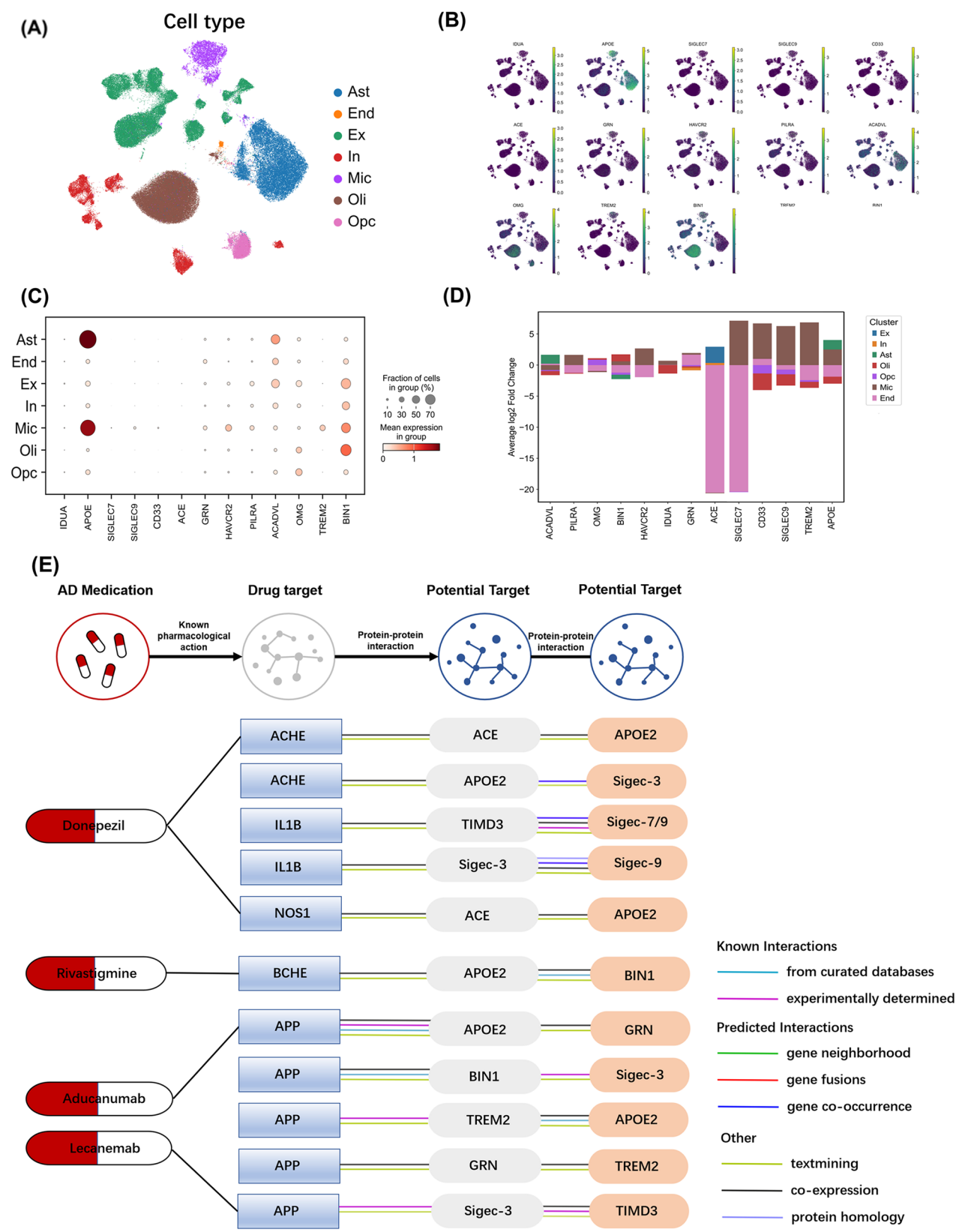


Fig. 3 Single-cell type expression in the brain tissue of AD patients for the coding genes of proteins identified by proteome-wide Mendelian randomization. **A** A total of 15 cell clusters and 7 cell types were identified. **B** and **(C)** Expression of protein-coding genes in each cluster. **D** Thirteen protein-coding genes were enriched in a cell type at an average Log₂FC > 0.25 and a P_{FDR} < 0.05. **E** Interactions between current AD medication targets and identified potential drug targets

well-established findings but also effectively identifies novel candidates for further investigation. Through comprehensive MR analysis of neuroimaging data, cognitive phenotypes, and biomarkers, we identified possible pathways associated with potential drug targets. The possible physiological mechanisms underlying the effects of IDUA (brain), APOE2 (brain), Siglec-7 (CSF) and Siglec-9 (CSF) on AD risk were further explored via mediation analysis. The methylation and transcription of these genes may be possible mechanisms for exploring the regulation of AD-related drug targets in the future. Additionally, our study revealed that the expression of these target genes was mostly enriched in microglia and astrocytes. These genetically predicted druggable targets may be associated with potential risks of nonepithelial cancer of the skin, retinal detachment, and metabolism-related diseases.

APOE isoforms are associated with several AD-related phenotypes, such as lipid risk factors and APP processing, which can be used to draw inferences on how it can be approached as a potential druggable target [11, 51]. Our integrated proteomic MR analysis revealed that APOE2 expression in brain tissue exerts widespread beneficial effects across multiple phenotypes, including hippocampal volume, cingulum-hippocampus MD values, global structural connectivity, and default mode network integrity. The APOE2 locus colocalized with AD risk signals—a finding that is consistently supported by multiple genomic databases—confirming its high priority as a therapeutic target. Mediation analysis further indicated that APOE2 may ameliorate AD risk and preserve the integrity of the hippocampal cingulate bundle partly by elevating CSF A β 42 levels. From a mechanistic perspective, APOE2 may confer protection against AD through multifaceted biological pathways beyond amyloid clearance [52]. In terms of lipid metabolism, APOE2 enhances lipid transport and cholesterol homeostasis, supporting synaptic integrity and neuronal membrane stability [53]. Its favorable lipid-binding properties facilitate efficient clearance of cytotoxic lipids and reduce lipid peroxidation, thereby attenuating neuroinflammation and oxidative stress [53]. With respect to immunomodulation, APOE2 appears to suppress proinflammatory pathways—particularly through interactions with IL1B—modulating microglial activation and shifting microglia toward a phagocytic, anti-inflammatory phenotype [54]. This effect may enhance A β clearance and reduce tau hyperphosphorylation. PPI analyses revealed that APOE2 engages with key players in AD pathogenesis, including APP, ACHE, BCHE, IL1B, NOX4, and NOS1. These interactions suggest the involvement of amyloid precursor protein processing, cholinergic signaling, oxidative stress regulation, and neuroinflammatory cascades. Specifically, connections with NOX4 and NOS1 play a role in mitigating reactive oxygen species production and

endothelial dysfunction, further supporting cerebrovascular health [55, 56]. Taken together, these findings position APOE2 as a multilevel protective mediator in AD, influencing amyloid pathology, lipid homeostasis, neuro-immune regulation, and redox balance.

The Siglec receptors are emerging as potential drug targets in AD, as they influence neuroinflammation, immune surveillance, and metabolic reprogramming. Previous proteomic studies have shown that, as a subtype of the Siglec receptor, Siglec-3 has the potential to be an AD-related drug target protein [57, 58]. From an immunological perspective, Siglec receptors act as checkpoint modulators that fine-tune neuroinflammatory responses. Siglec-3 promotes pro-inflammatory microglial activation, potentially accelerating A β accumulation and impairing phagocytosis [59]. Its interaction with IL1B, as revealed by PPI analysis, suggests a role in amplifying neuroinflammation through cytokine signaling. Meanwhile, Siglec-7 and Siglec-9—expressed on NK cells and microglia—may suppress immune surveillance via sialic acid-mediated inhibitory signaling [60]. This dampens NK cell cytotoxicity and microglial clearance of pathological proteins and senescent cells, facilitating p-tau propagation and synaptic damage. Metabolically, Siglec receptors engage in sialic acid-dependent metabolic reprogramming of immune cells [61]. By binding to sialylated glycans on neuronal or glial surfaces, Siglecs may alter energy metabolism and mitochondrial function in microglia, shifting them toward a dysfunctional phenotype that favors tau hyperphosphorylation and oxidative stress [61, 62]. Although currently classified as Tier 3 targets, Siglec-7 and Siglec-9 support a multipronged disease-modifying strategy for AD therapy through their involvement in innate immune regulation and tau pathology. These findings extend the previous associations of Siglec-3 with AD by identifying Siglec-7 and Siglec-9 as potential immunomodulatory candidates, suggesting a broader role of Siglec signaling in tau propagation.

Our results suggest that IDUA may reduce the risk of AD by modulating CSF levels of A β 42 and p-tau through its role in glycosaminoglycan (GAG) catabolism and lysosomal function. Unlike well-established AD risk genes, IDUA represents a novel association. Its involvement in GAG catabolism and lysosomal function provides a mechanistic link to A β and tau accumulation that has not been systematically explored in human genetic studies. Defects in IDUA cause mucopolysaccharidosis, and notably, the accumulation of A β and p-tau has been observed in mouse models of this disorder [63], suggesting a plausible mechanistic link between impaired GAG degradation and AD-related proteopathy. IDUA may influence AD pathogenesis through several pathways. By facilitating the breakdown of GAGs such as heparan sulfate—which is known to promote A β aggregation and tau

phosphorylation—IDUA can reduce the burden of pathological protein accumulation [64]. Additionally, by maintaining lysosomal function and autophagic flux, IDUA may increase the clearance of misfolded proteins and decrease neuroinflammation [65]. Improved lysosomal efficiency might also mitigate aberrant microglial and astrocyte activation, thereby contributing to preserved neuronal integrity. Although IDUA is currently classified as a Tier 4 target because of insufficient replication across databases, its biological relevance to protein clearance pathways and lysosomal function warrants further investigation.

Our study revealed that PILRA, TIMD3, OMGP, and ACADV are druggable protein targets identified from plasma proteomic analyses, which collectively highlight diverse pathogenic pathways spanning viral response, protein clearance, and lipid metabolism for potential therapeutic intervention in AD. We found that genetically predicted higher levels of the two soluble isoforms of PILRA (deltaTM and M14) in plasma were associated with a lower risk of AD. Mechanistically, PILRA may confer protection by limiting HSV-1 viral entry through reduced interaction with viral glycoprotein D, thereby attenuating neuroinflammation and A β accumulation triggered by herpesviral infection [66]. TIMD3, which binds to phosphatidylserine on apoptotic cells and interacts with amyloid precursor protein, may help regulate the clearance of cellular debris and modulate amyloidogenic processing [18]. OMGP, an oligodendrocyte myelin glycoprotein enriched in the CNS, was suggestively protective against AD and associated with improved reaction time, highlighting its potential role in preserving white matter integrity and cognitive processing speed [67, 68]. Conversely, elevated plasma ACADV levels correlated with worsened cognitive performance, possibly through disruptions in fatty acid oxidation and mitochondrial energy metabolism, leading to impaired neuronal function. Several plasma targets (PILRA, ACE, TREM2, BIN1, and GRN) have been previously reported in MR studies, strengthening their credibility [10, 11, 69]. These findings were extended by text mining: ACE may coregulate pathways with APOE and ACHE; BIN1 and TREM2 are strongly tied to amyloid plaque accumulation; and GRN supports lysosomal function and Notch signaling, facilitating the clearance of misfolded proteins [70]. We observed that higher plasma TREM2 levels were associated with a delayed age at onset of AD, potentially reflecting its role in enhancing microglial phagocytosis and promoting amyloid clearance. Tier assignment highlights robust targets such as PILRA with antiviral mechanisms, whereas tentative targets such as BIN1 are supported by weaker colocalization evidence. Collectively, these results highlight cell type-specific mechanisms—spanning viral response, lipid metabolism, myelination, and protein

clearance—through which these candidates may influence AD risk.

While recent proteomic studies have expanded the list of potential drug targets [10, 11, 58], our study stands apart by linking each target to causal genetic pathways and assessing their druggability potential. For instance, the analysis of APOE2 demonstrated how our integrative framework can elucidate the underlying mechanisms of genetic protection against AD. Our study further revealed that APOE2 in brain tissue may confer protection by increasing CSF A β 42 levels, thereby reducing the genetic risk of AD and improving neuroimaging metrics. Our findings also suggest that APOE2 is potentially coexpressed with the ACHE target of donepezil and the APP target of lucanemab and aducanemab, highlighting its promising druggability potential and offering specific mechanistic insights for drug development. This approach ensures that drug targets are not only identified but also mechanistically understood, making them more suitable for therapeutic intervention. Recent work by Ali et al. prioritized candidate genes on the basis of CSF proteomic profiling, providing direct evidence of protein alterations in the CNS [71]. In contrast, our integrative framework combined proteogenomic causal inference and MR across plasma, CSF, and brain pQTLs, alongside neuroimaging and cognitive data. Our framework successfully recapitulates well-established AD-associated proteins such as APOE and TREM2 while revealing novel candidates, including Siglec-7, Siglec-9, and IDUA. These findings not only extend previous proteomic observations but also provide complementary mechanistic insights into AD pathophysiology, highlighting potential targets for therapeutic intervention. These complementary perspectives highlight the need to integrate diverse proteomic strategies to comprehensively map AD biology and therapeutic opportunities.

Despite the inherent trade-off between resolution and robustness, proteomic MR capitalizes on existing large-scale GWAS resources and thereby circumvents additional experimental costs. More importantly, it enables discovery in settings where sample size cannot be feasibly expanded, such as rare diseases or hard-to-measure phenotypes, offering insights that traditional large-scale studies are unlikely to achieve [8, 9, 72]. By combining multitrait data, tissue-specific pQTLs, and annotations, proteomic MR improves the detection of causal signals beyond the reach of GWAS or proteomics alone [73, 74]. Evidence increasingly shows that therapeutic targets supported by human genetics are more likely to succeed in clinical development [75]. Proteomic MR has already contributed to this process by accelerating target prioritization and validation across diverse disease areas, with several findings subsequently corroborated in independent cohorts and clinical studies, underscoring